Smart Personalized Activity Routes for Kids (SPARK)

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Abstract and Project Objective

SPARK is an interactive tool that helps parents and caregivers **(HM Q4)** discover nearby activities for children and generates personalized daily schedules. By combining geographic data, family preferences, and development details, SPARK recommends activities aligned with a child's developmental priorities and a caregiver's practical lifestyle **(HM Q1).**

Current Practices and Literature Review

Today, parenting involves substantial cognitive labor, including planning family activities, coordinating schedules, and identifying opportunities for children's enrichment [1]. Research also shows that parents consider children's developmental and future-oriented needs when organizing their activities [2]. Existing tools largely address individual aspects of this process, such as scheduling or activity discovery, rather than bringing these considerations together into a coherent, developmentally informed activity plan **(HM Q2).**

Theme #1: Child Development and Caregiver Needs

Various studies highlight the benefits of different activities for children. Active play has been associated with cognitive benefits [3], sports participation with psychological and social benefits [4], and child-led art with personal and social development [5]. Outdoor play is shown to be critical for executive functioning, emotional resilience, and gross motor skill development [6]. Active play interactions can reduce overall familial stress, but these interactions tend to decline in quality and frequency when parents are stressed [7]. If stress becomes chronic for the parents, it can negatively impact decision-making, and lead to overly cautious or rigid practices [8]. Overall, these studies don't draw direct causal relationships between child development and activities, but our project can help mitigate parental stress by optimizing a well-balanced and practical schedule.

Theme #2: Neighborhoods and Child Friendly Environments

Research demonstrates that playful learning landscapes can incorporate opportunities for learning in everyday urban spaces. A new framework (focused on individual sites) measures the effects of playful environments in everyday spaces on children, families, and communities. [9]. Current research also shows that the "playability" of a neighborhood also has an effect on children's development [10]. There is also prior research conducted on the current state of urban design catered towards children, learning and community engagement [11]. However, there are some limits with current methods: They don't allow caregivers to access available activities in their areas, and in some cases, the frameworks are new.

Theme #3: Recommendation Systems, Data and Interface Design

Context-aware POI recommendation systems show that incorporating factors such as location, time, weather, distance, and user preferences can improve recommendations [12,13]. For SPARK, this means that relevant factors should be considered for activities based on child-specific factors such as age, interests, developmental priorities, activity type, as well as practical constraints such as time and distance. High-quality geographic data is essential for this approach. OpenStreetMap provides an openly available source of POI data, but its completeness, accuracy, and timeliness vary [14,15]. The interface will combine map-based activity discovery with scheduling, building on research showing the importance of family calendars [16] and task-focused map design [17].

The Innovative Potential of PARK & It's Impact

SPARK is an innovative visualization that combines automated agenda generation, an interactive map with relevant local POIs and simple, yet meaningful customization parameters (age, child

development priorities, general preferences). Computationally, SPARK combines POI data from OpenStreetMap (using SQLite for data storage) with child activity focused Feature Engineering that takes relevant POI data like "playground" and maps it to activities like "ball games" and "free play", which in turn have different scores on child development factors like "creativity" and "social skills". On top of this, SPARK features an optimization engine that finds the best combination of activities, given the selected parameters and a rudimental routing algorithm to map routes between activities and complement the optimized activity agenda with a map view that shows activity locations and travel paths between them. We expect this new type of interactive visualization to be successfully adopted by caregivers because it addresses the desire to streamline the process of planning child activities, taking stress out of the process while offering a wider variety of activities that incorporate key child development factors (**HMQ3**).

If successful, SPARK has an impact on caregivers by reducing stress and cognitive effort involved in planning child activities while addressing desires to incorporate a more varied set of activities optimized for age-appropriate development priorities. For children, SPARK helps expose them to activities that help them progress towards development milestones relevant for their age (social skills, confidence, creativity, etc.) and gain exposure to a greater variety of activities in their neighborhood that connect them to their local community. For both caregivers and children, user studies, focus interviews and questionnaires will enable success measurement (**HMQ5**).

Risks, Payoff and Cost

A key risk is that SPARK requires the integration of various data sources and models within a relatively short project time frame. Another risk is that caregivers in rural or very residential neighborhoods may not have a significant variety of relevant POIs and activities in their vicinity, making the tool less appealing in such areas. Potential payoffs include a positive impact on child development through more thoughtful activity planning and reduced stress of caregivers in planning child activities. If the initial version of SPARK is widely adopted, there may also be monetization potential via targeted advertisements on the map and premium features (**HM Q6**)

The initial version of SPARK will focus on limited geographical areas, take advantage of open-source data sets, models and platforms, and be built by student data and computer scientists as part of their coursework. This results in zero monetary costs (**HM Q7**).

Project Plan and Success Criteria

Effort Distribution: *All team members are expected to contribute a similar amount of effort throughout the project.*

The project will be completed within 11 weeks (**HM Q8**). Our "midterm" goal to track success entails completing the first 4 activities in the Plan of Activities (PoA) below. Final success is to complete all activities in the PoA. Progress will be measured via weekly group review of activity statuses in the PoA ***(HM Q9).***

Plan of Activities

Activity	Owner(s)	Timeline	Status
Literature review & project design	All	Sep 21 – Oct 4	**Done**
Identification of data sets	Max, Manoj, Shan	Sep 28 – Oct 12	**Done**
Data preparation and model selection	Max, Manoj, Shan	Oct 5 – Oct 25	**In Progress**
Develop initial models and UI	Shan, Manoj	Oct 25 – Nov 15	**Not Started**
Integrate the models into the UI	Shan, Manoj	Nov 9 – Nov 22	**Not Started**
Model evaluation and testing	Jay, Katherine	Nov 16 – Nov 29	**Not Started**
Final report and poster development	All	Nov 23 – Dec 3	**Not Started**

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